CDA 106
Numerical Methods for Data Science

Dr Prashant Kumar Srivastava

IIT Patna

pksri@iitp.ac.in

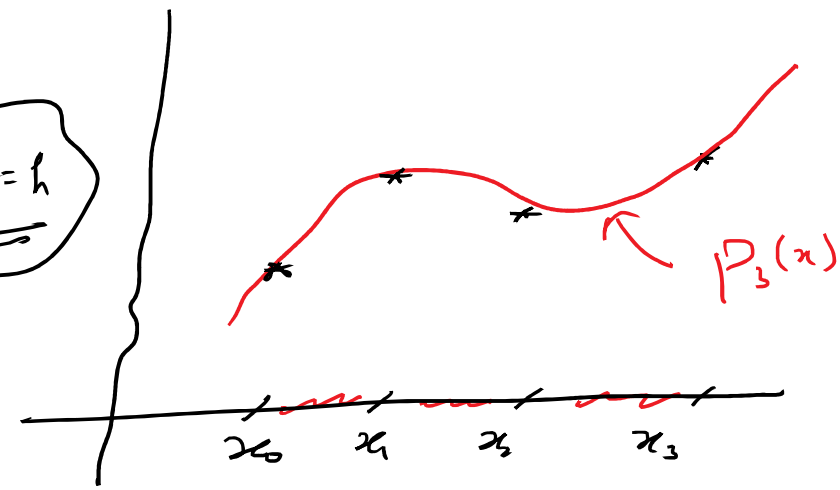
LECTURE-7 February 20, 2026

Numerical Differentiation

x	x_0	x_1	x_2	x_3
$y = f(x)$!!	$f(x_0)$	$f(x_1)$	$f(x_2)$	$f(x_3)$

h h h

$x_{i+1} - x_i = h$



Can we find $f'(x)$??

→ What if we fit the data to a "polynomial"

→ Newton-Gregory "forward difference" interpolation formula.

→ $\Delta f(x_i) = f(x_{i+1}) - f(x_i) \quad \forall i$

$= f(x_i + h) - f(x_i)$

Newton-Gregory Forward difference Interpolation formula

$$p_n(x) = f(x_0) + \frac{x-x_0}{h}\Delta f(x_0) + \frac{(x-x_0)(x-x_1)}{2!\,h^2}\Delta^2 f(x_0) + \frac{(x-x_0)(x-x_1)(x-x_2)}{3!\,h^3}\Delta^3 f(x_0) + \cdots + \frac{(x-x_0)(x-x_1)\cdots(x-x_{n-1})}{n!\,h^n}\Delta^n f(x_0)$$

- n th- degree polynomial with "$n+1$" points

Example

$h = 1$

x	3 (x_0)	4 (x_1)	5 (x_2)	6 (x_3)
$f(x)$	6 ($f(x_0)$)	24 ($f(x_1)$)	60 ($f(x_2)$)	120 ($f(x_3)$)

Construct a forward difference Table

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
3 (x_0)	6 ($f(x_0)$)	18 ($\Delta f(x_0)$)	18 ($\Delta^2 f(x_0)$)	6 ($\Delta^3 f(x_0)$)
4	24	36	24	
5	60	60		
6	120			

From Newton Gregory - - -

$$p_3(x) = f(x_0) + \frac{x-x_0}{h}\Delta f(x_0) + \frac{(x-x_0)(x-x_1)}{2!\,h^2}\Delta^2 f(x_0) + \frac{(x-x_0)(x-x_1)(x-x_2)}{3!\,h^3}\Delta^3 f(x_0)$$

$$= 6 + \frac{x-3}{1}\times 18 + \frac{(x-3)(x-4)}{2!\,(1)^2}\times 18 + \frac{(x-3)(x-4)(x-5)}{3!\,(1)^3}\times 6$$

$$= 6 + 18x - 54 + 9x^2 - 63x + 108 + x^3 - 12x^2 + 47x - 60$$

$$\boxed{p_3(x) = x^3 - 3x^2 + 2x}$$

Note carefully

This polynomial $p_3(x) = x^3 - 3x^2 + 2x$ satisfy the given data:

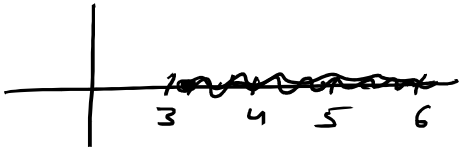
$$p_3(x_0) = f(x_0)$$
$$p_3(x_1) = f(x_1)$$
$$p_3(x_2) = f(x_2)$$
$$p_3(x_3) = f(x_3)$$

$$p_3(3) = 3^3 - 3\times 3^2 + 2\times 3 = 6$$

$\vdots$ you can verify

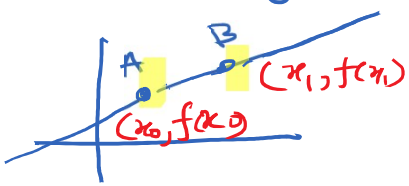
Now, suppose if I want $f'(3) = ?$

$$f'(3) = p'(3) = \left[3x^2 - 6x + 2\right]_{x=3}$$
$$= 27 - 18 + 2$$
$$\boxed{f'(3) = 11}$$



Some fact

Q. How many min points I will require to draw a straight. $p_1(x) = a + bx$



$$\left.\begin{aligned} p_1(x_0) &= f(x_0) \\ p_2(x_1) &= f(x_1) \end{aligned}\right\}$$

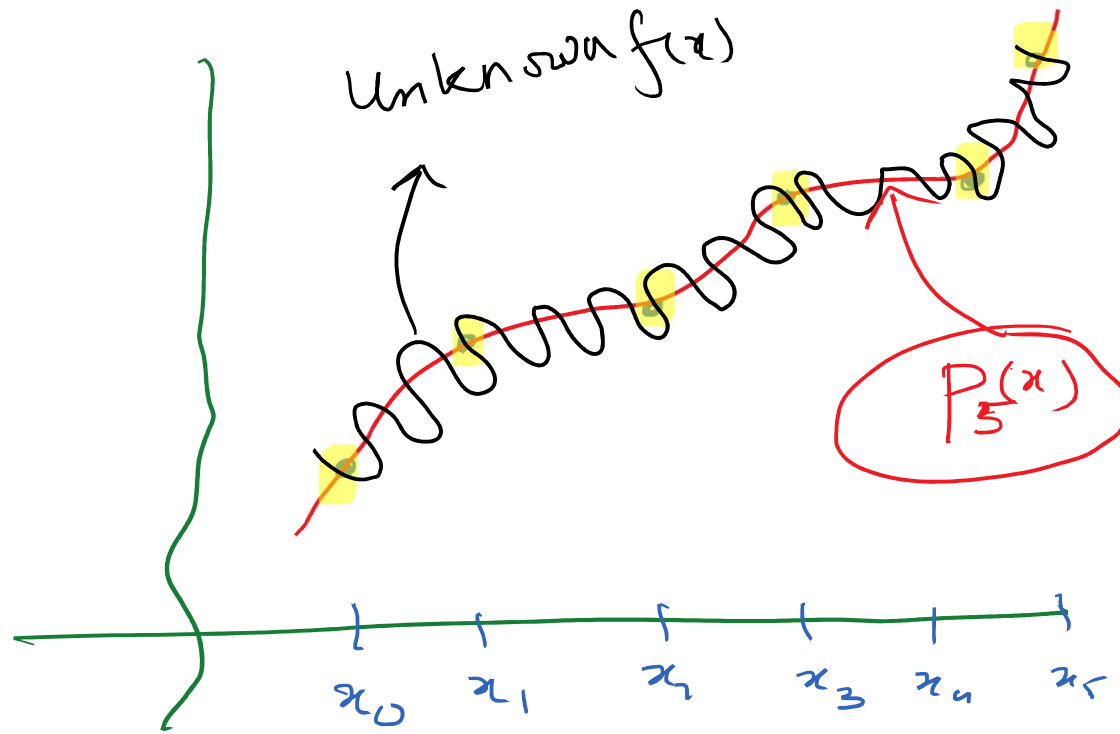
2 eqn in a & b

Q. How many points I shall now need to draw

$p_2(x) = ax^2 + bx + c$ // 3 points

Q. How many points do I need to draw a "$p_n(x)$" - n th deg. polynomial

— "$n+1$"

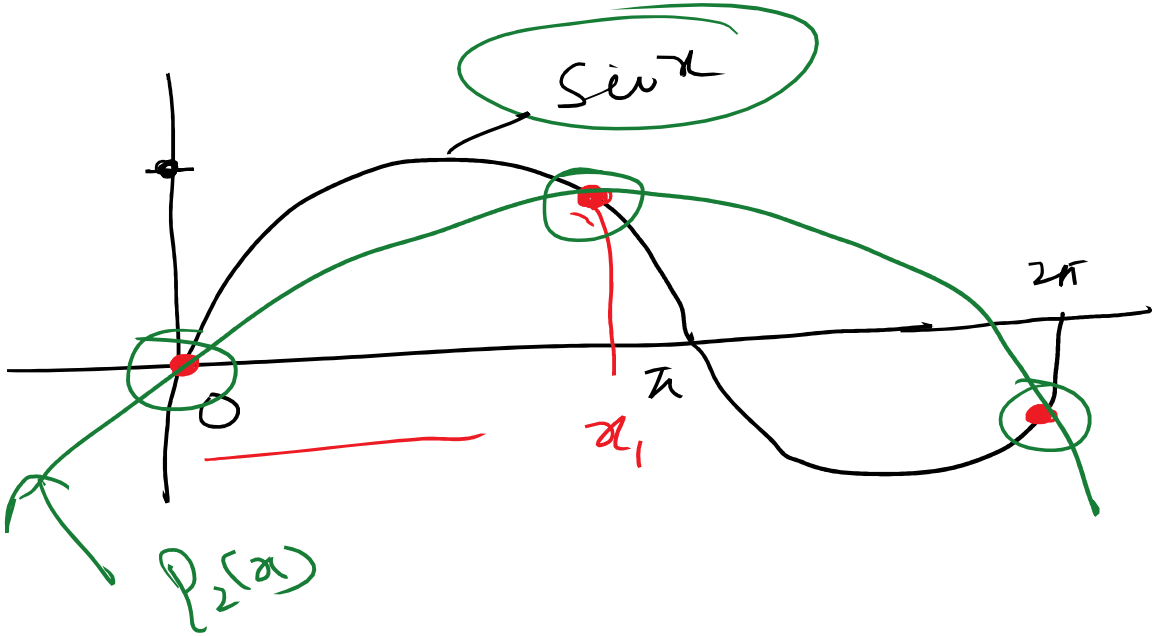


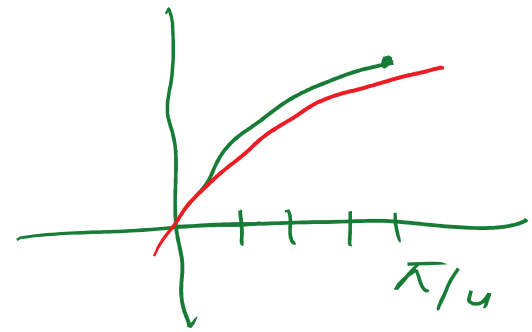
(equidistant)

$P_5 \approx f(x)$ best possible

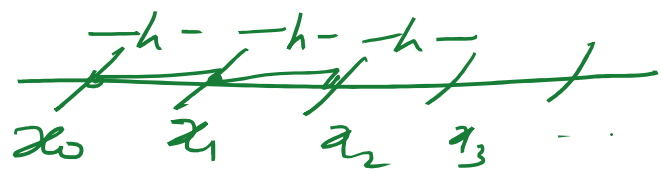
x_0	x_1	x_2	x_3	x_4	x_5
$f(x_0)$	$f(x_1)$	$f(x_2)$	$f(x_3)$	$f(x_4)$	$f(x_5)$

[[Unknown function & fitted polynomial "match" at data points





My fitted polynomial is best approximation of unknown <u>function</u>



$x_1 = x_0 + h$

$x_2 = x_0 + 2h$

→ Having known the $\boxed{\text{polynomial } (P_n(x))}$, we can always find $f'(x)$

$$\left(f'(x) = P_n'(x)\right)$$

<u>Que</u> Do we <u>necessarily</u> need to find $(P_n(x))$ <u>polynomial</u> to find derivative?

"<u>NO</u>" We can directly calculate

$$f'(x_*) = ?$$

Newton-Gregory Forward difference Interpolation formula

$(x_0, f(x_0)) \ldots (x_n, f(x_n)) \leftarrow n+1$ ~~pair~~ data points (equidistant)

$x_{i+1} - x_i = h$

$$p_n(x) = f(x_0) + \frac{x-x_0}{h}\Delta f(x_0) + \frac{(x-x_0)(x-x_1)}{2!\,h^2}\Delta^2 f(x_0) + \frac{(x-x_0)(x-x_1)(x-x_2)}{3!\,h^3}\Delta^3 f(x_0) + \cdots + \frac{(x-x_0)(x-x_1)\cdots(x-x_{n-1})}{n!\,h^n}\Delta^n f(x_0)$$

Consider $q = \frac{x-x_0}{h}$; $\frac{x-x_1}{h} = \frac{x-x_0-h}{h} = q-1$

$\frac{x-x_2}{h} = \frac{x-x_0}{h} - 2 = q-2, \ldots, \frac{x-x_{n-1}}{h} = q-(n-1)$

$x_1 - x_0 = h$

$x_1 = x_0 + h$

$x_2 - x_0 = 2h$

$x_2 = x_0 + 2h$

$p_n(x) \simeq f(x)$

x is any point between $x_0 - x_n$

$$f(x) = f(x_0) + q\Delta f(x_0) + \frac{q(q-1)}{2!}\Delta^2 f(x_0) + \cdots + \frac{q(q-1)(q-2)\cdots(q-(n-1))}{n!}\Delta^n f(x_0)$$

$q = \frac{x-x_0}{h}$

Notice carefully

$$f(x) = f(x_0) + q\,\Delta f(x_0) + \frac{q(q-1)}{2!}\Delta^2 f(x_0) + \frac{q(q-1)(q-2)}{3!}\Delta^3 f(x_0) + \cdots = F(q)$$

In order to find $f'(x)$ — we need to note that RHS of the above formula is a function of $q = \frac{x-x_0}{h}$

So that $f'(x) = \frac{df(x)}{dx} = \frac{d}{dq}F(q) \cdot \frac{dq}{dx}$ $\quad \left| \frac{dq}{dx} = \frac{1}{h} \right.$

$$\Rightarrow f'(x) = \frac{1}{h}\left[\Delta f(x_0) + \frac{2q-1}{2!}\Delta^2 f(x_0) + \frac{3q^2-6q+2}{3!}\Delta^3 f(x_0) + \cdots\right]$$

$$f''(x) = \frac{d}{dx} f'(x)$$

$$= \frac{d}{dq}\left[\quad '' \quad\right] \cdot \frac{dq}{dx}$$

$$f''(x) = \frac{1}{h^2}\left[\Delta^2 f(x_0) + (q-1)\Delta^3 f(x_0) + \cdots\right]$$

$y = f(x)$

$= f(\phi(t))$

$= \psi(t)$

$x = \phi(t)$

$$\boxed{\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}}$$

Chain rule

$y = \sin t^2$ " $\equiv$ $\sin x$, $x = t^2$

$$\frac{dy}{dt} = \frac{d(\sin x)}{dx} \times \frac{d(t^2)}{dt} = \cos x \cdot 2t$$

$$= 2t \cdot \cos t^2$$

Example

x	3	4	5	6
$f(x)$	6	24	60	120

$\Delta f(x_0) = 18$

$\Delta^2 f(x_0) = 18$

$\Delta^3 f(x_0) = 6$

Q.1. find $f(3.5) = ?$

$h = 1$, $x_0 = 3$

Q.2 find $f'(3) = ?$

Ans 1: $x = 3.5$ $\quad q = \frac{x - x_0}{h} = \frac{3.5 - 3}{1} = 0.5$

$$f(3.5) = f(x_0) + q\,\Delta f(x_0) + \frac{q(q-1)}{2!}\Delta^2 f(x_0) + \frac{q(q-1)(q-2)}{3!}\Delta^3 f(x_0)$$

$$= 6 + 0.5 \times 18 + \frac{0.5(-0.5)}{2} \times 18 + \frac{0.5(-0.5)(-1.5)}{6} \times 6$$

$$= 6 + 9 - 2.25 + 0.375 = 13.125$$ Ans.

Ans 2

$$\boxed{x = 3}$$

$$q = \frac{x - x_0}{h} = \frac{3-3}{1} = 0$$

$$f'(3) = \frac{1}{h}\left[\Delta f(x_0) + \frac{2q-1}{2}\Delta^2 f(x_0) + \frac{3q^2-6q+2}{6}\Delta^3 f(x_0) + \cdots\right]$$

$$= \frac{1}{1}\left[18 + \frac{2\times 0 - 1}{2}\times 18 + \frac{3\times 0^2 - 6\times 0 + 2}{6}\times 6\right]$$

$$= 18 - 9 + 2 = 11$$

Ans.

In brief.

① We found a method (Newton-Gregory) to fit a polynomial with a given data.

② Deg. ⟶ no of points

③ forward difference table (useful)

④ We used F.D.T. to find polynomial & values of the function at intermediate points

⑤ $q = \frac{x - x_0}{h}$ & we found $f(x)$ & $f'(x)$ & $f''(x)$ formulae.

"Interpolation"

x_0 x_n